

SAUTRIK ROY

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EDUCATION

SRM Institute of Science and Technology (SRM IST)

Chennai, India

Bachelor of Technology in Computer Science and Engineering (Core) — **CGPA: 8.77 / 10.0**

Expected May 2029

Relevant Coursework: Data Structures & Algorithms, Object-Oriented Programming (C++), Web Technologies, Database Systems (DBMS), Operating System Fundamentals.

TECHNICAL SKILLS

Languages:	C, C++, JavaScript (ES6+), TypeScript, Java, Python, SQL, HTML5, CSS3
Frontend:	React, Next.js, Vite, Tailwind CSS, Responsive Web Design, Web Audio API, Canvas API
Backend & DB:	Node.js, Express.js, RESTful APIs, WebSockets, Supabase, PostgreSQL, MySQL
Tools & Workflow:	Git, GitHub, GitHub Actions, Postman, Linux / Bash, Vercel, VS Code
Core Competencies:	Data Structures & Algorithms, OOP, W3C Web Standards, Relational Database Design

EXPERIENCE & COMMUNITY LEADERSHIP

Microsoft Learn Student Ambassador (MLSA)

Jan 2025 – Present

Student Ambassador — Campus Community

SRM IST, Chennai

- Organized campus technical workshops and peer learning sessions on Git/GitHub version control, software development fundamentals, and modern developer tooling.
- Mentored 40+ first-year computer science students on environment configuration, web development foundations, and collaborative coding practices.

GirlScript Summer of Code (GSSoC)

2025

Open Source Contributor

Remote

- Contributed to active open-source repositories by addressing frontend UI issues, refining component responsiveness, and submitting tested pull requests in React and JavaScript.
- Collaborated with community maintainers following structured Git branch workflows and participating in peer code reviews.

TECHNICAL PROJECTS

QRCraft Studio — High-Precision Vector QR Generator & Visual Studio

Live Demo | GitHub

React, Vanilla CSS, Vite, HTML5 Canvas, Web Audio API, Vercel

2026

- Built a 100% client-side QR generation suite with zero server latency, supporting 300 DPI PNG raster and infinite-resolution SVG vector exports rendered directly in-memory.
- Implemented an automated W3C WCAG 2.1 relative luminance algorithm to audit foreground-to-background contrast ratios in real time, alerting users to low-scannability combinations.
- Engineered custom device mockup stages, a synthesized Web Audio API sound effect system, and payload formatters conforming to RFC 6068 (Mailto), RFC 3966 (Tel), and Wi-Fi standards.

Sentinel — Telehealth & Mental Readiness Assessment Platform

SIH Top 100 | GitHub

React, Node.js, Express.js, Supabase, WebRTC, PWA, IndexedDB

2025 – 2026

- Developed a full-stack health evaluation portal for defense personnel wellness tracking, shortlisted among the Top 100 teams out of 130 in the SRM Smart India Hackathon internal round.
- Integrated WebRTC peer-to-peer audio/video streaming for direct remote consultation sessions with low latency.
- Implemented an offline-first PWA architecture using IndexedDB to cache assessment questionnaires, allowing full data entry during network outages with automated sync upon reconnection.

Loop Beyond Limits — Ad-Free Music Streaming Web Application

GitHub

React, TypeScript, Node.js, Supabase, PostgreSQL

2025

- Designed and built a full-stack music streaming web player featuring audio playback controls, search filtering, and customizable user playlists.
- Configured Supabase PostgreSQL authentication with Row-Level Security (RLS) policies to protect user library data and manage active session states securely.
- Crafted a responsive, dark-mode-first user interface with smooth CSS transitions and cross-device audio stream compatibility.

Quizzify — Automated Quiz Generation & Assessment Tool

GitHub

C++, Google Gemini API, Object-Oriented Programming, REST APIs

2025

- Constructed a C++ application that extracts text from documents and connects to the Google Gemini REST API to dynamically generate formatted multiple-choice quizzes.
- Utilized object-oriented design patterns (inheritance, polymorphism, encapsulation) to maintain separate Practice and Timed Test evaluation modes with shared scoring logic.

HONORS & CERTIFICATIONS

- Smart India Hackathon (SIH) 2026:** Shortlisted among Top 100 Teams (out of 130 teams / 780 participants) at SRM IST internal selection for Project Sentinel.
- Microsoft Certified:** Azure Fundamentals (AZ-900).
- Academic Merit:** Maintained an 8.77 / 10.0 CGPA throughout B.Tech Computer Science & Engineering coursework.